

MARSHALL BRUNER

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EXPERIENCE

Ball Aerospace (DoD Clearance)

RF Engineer

Jun 2021 — Jul 2023

Broomfield, CO

- Designed and debugged multiple complex, mission-critical RF boards under strict time constraints
- Schematic capture and layout of RF T/R systems (RF, digital, power), antenna range test equipment, component testing coupons, etc.
- Built data visualization and analysis tools in **Python**
- Built an embedded webpage used to control/monitor phased array electronics
- High sensitivity testing (noise, high power, etc.) of electronics

Colorado State University

Graduate Research Assistant — Thesis

Jan 2021 — May 2022

Fort Collins, CO

- **Design, schematic, and layout** of a modular, X-band, FMCW drone radar
- **Signal processing / data visualization** for an FMCW radar board
- Presented the work at the IEEE International Phased Array Symposium

CHILL Radar Lab

Lab Assistant

May 2020 — May 2021

Greeley, CO

- Built a real-time radar display to easily display and animate radar data
- Built a server to host and display real-time and historical **geospatial radar data** using **PostgreSQL**, Flask, **Python**

EDUCATION

Colorado State University

M.S. Electrical Engineering — GPA: 3.9

Jan 2021 — May 2022

Fort Collins, CO

Thesis - Design, Deployment, and Cost Considerations for DARMA; A Low-Cost and Lightweight FMCW Radar

Colorado State University

B.S. Electrical Engineering — GPA: 3.47

Aug 2017 — May 2021

Fort Collins, CO

PROJECTS

Algorithmic Stock Trading Server

Aug 2022 — Present

- Task scheduling system (backfilling data into **PostgreSQL** database, computing custom indicators, run entry / exit strategy)
- Containerized using docker
- Command-line interface for checking status, getting recent trades, etc.
- Analysis / forecasting using **Pandas**, **Numpy**, **PyTorch**
- Dashboard using **Plotly Dash** and a **Flask** API

FMCW Radar Bootcamp

Mar 2023

- **Developed signal processing software** and a presentation for an FMCW radar to present at the 2022 AESS Radar Bootcamp

5G Electronics Senior Design Project

Jun 2020 — May 2021

- Simulated, assembled, and measured a full 5G receiver system
- Created teaching documentation for a 5G transmitter and receiver for FR1 and FR2 frequency bands

SKILLS & ABILITIES

Languages: **Python**, Rust, C, Bash, **SQL**, Matlab, NuShell, JavaScript

Technologies / Tools: Git, $\text{\LaTeX}$, Linux, Docker, **PostgreSQL**, **Plotly**, Flask, Celery

Communication: Worked with customers through pre- and post-sales phases, Experience Presenting PDR/CDRs, Technical Presentation Formatting